ANSIK SINGH TOMAR

Gwalior, Madhya Pradesh, India Portfolio
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EDUCATION

Indian Institute of Information Technology Guwahati **2024 – 2028**
Bachelor of Technology in Computer Science and Engineering; CGPA: 9.27 *Guwahati, Assam, India*

Greenwood Public School Gwalior **2021 – 2023**
Intermediate (Class XII); Percentage: 93.8% *Gwalior, MP, India*

Army Public School Gwalior **2014 – 2021**
High School; Percentage: 93.6% *Gwalior, MP, India*

SKILLS

- **Languages:** C, C++, Java, Python, JavaScript, SQL
- **Frontend:** React, Vite, Tailwind CSS, HTML5, CSS3, Responsive UI
- **Backend:** Node.js, Express, FastAPI, REST APIs, JWT Authentication
- **Database & Cloud:** MongoDB, MySQL, Firebase, Cloudinary
- **Tools:** GitHub, Git, Visual Studio Code, Postman, Arduino IDE, Vercel
- **Concepts:** DSA, DBMS, OOP, Computer Architecture, Operating System, Computer Networks

EXPERIENCE

Internship – 2 Months | Full Stack Web Developer Intern | Technology Business Incubator, GEU Dehradun

- Engineered an end-to-end web portal for agricultural farmer collectives in the Himalayan region to connect organic growers directly with bulk buyers.
- Implemented role-based access control (RBAC) and JWT authentication with Node.js and Express for distinct Farmer, Collective, and Admin dashboards.
- Designed normalized MongoDB schemas and transactional workflows to handle crop listings, deal requests, and pickup scheduling without data conflicts.
- Integrated Cloudinary for media verification uploads and configured automated Nodemailer notifications for pickup schedules and payment updates.

PROJECTS

FarmFresh – Full-Stack Agricultural Marketplace & Supply Chain Platform

Tech Stack: React, Node.js, Express, MongoDB, Tailwind CSS, Groq API, Cloudinary, Vercel

- Developed a dual-sided marketplace enabling farmer groups to negotiate crop deals, schedule batch pickups, and maintain transparent ledgers.
- Architected a zone-based pickup scheduler with vehicle capacity constraints, live driver assignments, and immutable UTR payment records.
- Integrated Groq LLaMA 3.1 AI advisor (*FarmAssist*) to provide contextual crop management and soil guidance via a responsive chat interface.
- Configured stateless JWT authentication with automated Axios token-refresh interceptors and resilient error handling on Vercel deployment.

Insight Forge – Natural Language Data Analytics Platform

Tech Stack: React, FastAPI, Python, Groq API, SQLite, Tailwind CSS, Vercel, Render

- Developed a natural language analytics platform that transforms plain-English queries into executed SQL operations and visual insights.
- Built a 4-agent LLM pipeline (Table Selector, SQL Generator, Execution Engine with auto-fix retry, and Insight Generator) using Groq API.
- Implemented conversational session memory, automatic chart type selection (Recharts), and custom CSV dataset upload with schema inference.

Gesture and Voice-Controlled Robotic Arm

Tech Stack: Python, OpenCV, MediaPipe, Azure Speech Services, Arduino Uno, PyFirmata

- Built a working prototype of a 5-fingered robotic hand actuated in real time via dual input modes: hand-gesture tracking and spoken commands.
- Calculated 21 3D hand-landmark coordinates using MediaPipe and OpenCV, mapping joint angles to 5 SG90 servo motors via PyFirmata.
- Integrated Azure Cognitive Speech Services to enable voice-based control supporting English and Hindi spoken commands for preset grip routines.

ACHIEVEMENTS

- Secured **98.5 percentile** in JEE Mains (among 1.4+ million candidates nationwide)
- Received **ESSA** (Education Scholarship Scheme for Army Personnel) Merit Scholarship for academic excellence
- Academic Standing: Maintaining a **9.27 CGPA** in B.Tech CSE at IIIT Guwahati